https://doi.org/10.1038/s41591-026-04433-3

Genetic–exposome interactions and aging clocks in dementia: the ReDLat2 initiative

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Neither genes nor the environment should be studied in isolation in brain health and dementia research. Across Alzheimer's disease and frontotemporal lobar degeneration, risk and resilience emerge from the interactions of inherited susceptibility, biological embeddings, and exposure to environmental factors accumulated across the lifespan (the exposome)[1,2]. Yet the field still relies disproportionately on populations of predominantly European ancestry living in relatively homogeneous environmental settings, which limits both discovery and translation. This gap is accentuated for Latin American populations, in which the dementia burden is high, genetic admixture is complex, environmental adversity is heterogeneous, and access to diagnosis, biomarkers and trials remains deeply unequal.

Two main gaps have constrained progress. First, gene–environment research in dementia has rarely incorporated the exposome in a harmonized and biologically meaningful way[2,3]. Social determinants of health, socioeconomic status, inequality and democracy, violence, nutrition, physical factors (for example, pollution, lack of green spaces, water quality, drought severity and agrochemicals), and country-level conditions are often treated as covariates rather than as interacting drivers of vulnerability and resilience[1,4–7].

Second, although there is evidence that both genetics and the exposome influence accelerated aging (vulnerability) and delayed aging (resilience), these have not been systematically integrated[1,3]. Aging clock research uses estimates of age based on various measures (organ, brain, behavioral, epigenetics and proteomics) and contrasts those against chronological age. Aging clocks offer a framework for integrating genetic and environmental influences into tractable phenotypes[2,3]. Thus far, aging clocks have been studied largely as isolated prediction tools rather than as integrative phenotypes that capture how genetic susceptibility and real-world exposures impact disease risk. In under-represented populations, these limitations are amplified by small samples, single-country designs, weak family-based discovery, poor ancestry resolution and insufficient deep phenotyping.

These barriers prevent precision brain health from predicting real-world phenotypes and related mechanisms[2]. There is still too little that is known about how common polygenic burden, rare pathogenic variants, founder effects, local ancestry and protective variants interact with adverse or protective exposomal conditions to shape disease onset, phenotype severity and biological aging. There are also no scalable frameworks for determining whether accelerated aging signals in the brain, behavior, cognition, blood biomarkers or epigenetic profiles indicate shared pathways of vulnerability or resilience.

The renewal of the Multi-Partner Consortium to Expand Dementia Research in Latin America (ReDLat2) was designed to address these barriers (Table 1; full list of members, Supplementary Information). Recently funded by the US National Institute on Aging, ReDLat2 builds on its predecessor, ReDLat1 (ref. 8), and will establish a deeply phenotyped, environmentally embedded and genomically informed dementia research program focused on Latin America and the USA. Spanning 13 centers in 7 countries and involving around 90 researchers and staff, the consortium will integrate Argentina, Brazil, Chile, Colombia, Mexico, Peru and the USA into a common scientific infrastructure for equitable precision dementia research.

ReDLat1 demonstrated the feasibility and value of this approach. It established harmonized protocols, centralized data management, dedicated working groups, a harmonized plasma biobank, and a robust multi-country legal and scientific infrastructure[8]. With more than 4,000 participants[8], the consortium generated hundreds of publications and produced early evidence that the exposome shapes dementia phenotypes in Latin American cohorts, while also identifying pathogenic variants, founder-related genetic effects and ancestry-linked genomic structures[9]. It also developed computational approaches to aging clocks and the exposome[4–6] capable of handling multimodal heterogeneity across sites, devices and populations, laying the groundwork for the next phase.

ReDLat2 will extend this platform by combining ReDLat1 with 3,000 newly recruited participants, including individuals with Alzheimer's disease or frontotemporal lobar degeneration and healthy control participants, as well as relatives from large multiplex families. Its central hypothesis is that dementia phenotypes are shaped by the interactions of exposome adversity, ancestry-informed genetic risk and multimodal aging clocks. To test this, the project integrates three linked aims.

The first aim examines how the exposome (social determinants of health, socioeconomic indicators and country-level measures of social and physical adversity) interacts with ancestry and polygenic risk to shape multimodal phenotypes across Latin America and non-Latino US cohorts. The second aim will characterize rare causal variants, protective variants, local ancestry effects and polygenic scores tailored for Latino populations, through case–control and family-based whole-genome sequencing in cross-sectional and longitudinal designs, including short-read and selected long-read sequencing. The third aim will develop multi-clock models of accelerated and delayed aging using machine learning, deep learning[6] and biophysical whole-brain modeling[10] to test how external parameters linked to environment or ancestry modulate disease differences across region, sex and diagnosis.

Large Latin American families with founder effects remain underutilized for gene discovery and therapeutic target nomination[9]. Polygenic scores tailored for Latino populations remain underdeveloped. Plasma biomarkers are increasingly central to trial readiness, but access, validation and deep phenotyping remain uneven in the region. ReDLat2 may help refine risk stratification, improve biomarker interpretation, expand equitable trial readiness and provide a template for globally relevant precision medicine.

Across the consortium, we will integrate validated exposome inventories, plasma biomarkers, proteomics, epigenetics, whole-genome

Table 1 | ReDLat2 framework of genetic–exposome interactions in dementia research

Barrier	ReDLat2 framework	Expected advance
Dementia research under-represents Latin American populations and relies on relatively homogeneous populations.	A multi-country, harmonized platform spanning Latin America and a non-Latino US comparison cohort within a shared protocol and scientific infrastructure	More globally relevant, heterogeneity-sensitive dementia evidence
Social and physical exposomes are ignored or often treated as covariates rather than as mechanistic drivers of disease.	Exposome integration of social determinants and aggregate-level measures, including physical and social adversities across cohorts	Tests of how exposomes shape dementia phenotypes and interact with genetic risk
Gene–environment interactions remain poorly characterized in dementia, particularly in under-represented populations.	Modeling of exposome, ancestry, polygenic risk and multimodal phenotypes, rather than treating genes and environment separately	Mechanistic accounts of vulnerability and resilience in real-world exposures
The genetic architecture of Latino families remains insufficiently characterized, including rare causal variants, founder effects, local ancestry and protective variants.	A combined family-based and case–control genomics platform using short-read whole-genome sequencing of participants and selected long-read sequencing in unresolved families	Discovery of causal, protective and ancestry-linked variants relevant to dementia in Latinos
Polygenic risk scores derived from European datasets do not generalize well to admixed Latino populations.	Polygenic analyses tailored for Latino populations, combined with local ancestry and tri-continental admixture modeling	Improved calibration and transferability of genetic risk
Biomarker access, harmonization and validation remain uneven across Latin America.	A cross-site plasma biomarker platform, including known markers of Alzheimer's disease and neurodegeneration, along with proteomics	Greater diagnostic precision, biomarker access and regional trial readiness
Brain aging models rarely incorporate exposome and genetic burden within the same framework.	A brain aging clock that uses MRI, machine learning and biophysical modeling, explicitly modulated by exposome and genetic risk	Diversity-sensitive and mechanistically informed phenotypes in dementia
Aging clocks are commonly used as isolated descriptive tools, rather than as integrative phenotypes.	Integration of complementary clocks, including brain, biobehavioral, speech, epigenetic, proteomic and ATN-related aging signatures	Systems-level framework that links accelerated and delayed aging to vulnerability and resilience
Clinical trials often lack inclusive, deeply characterized Latino cohorts and a coordinated regional infrastructure.	Biomarker-, genomics- and family-informed recruitment within a harmonized consortium	Equitable trial-ready cohorts and foundations for future therapeutic studies

MRI, magnetic resonance imaging; ATN, amyloidosis (A), tau pathology (T) and neurodegeneration (N).

sequencing, admixture and local ancestry analyses, natural speech analysis, circadian imbalance, neuroimaging, and multimodal computational modeling within harmonized recruitment and phenotyping pipelines.

The scientific relevance of this approach extends beyond Latin America. By explicitly modeling heterogeneity rather than averaging it, ReDLat2 offers a route for connecting real-world exposures and phenotypes with dementia biology[2,4–6]. It also brings together open science, secure data sharing, training pipelines and multi-country harmonization across diverse populations.

Most importantly, it will test a central hypothesis with broad relevance: that the interaction of genetic ancestry and environmental conditions is a principal source of variance in how aging and neurodegeneration unfold. If successful, ReDLat2 will help redefine aging clocks as integrative readouts of gene × environment interactions and reposition socio-biological heterogeneity as a scientific underpinning for precision brain health.

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Published online: 28 May 2026

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Funding

ReDLat2 is supported mainly by the US National Institute on Aging (R01 AG057234). A.I. is supported by grants from ReDLat, including the Fogarty International Center, the US National Institutes of Health (NIH), the US National Institute on Aging (R01 AG057234, R01 AG075775, R01 AG21051, R01 AG083799, CARDS-NIH and R01 AG057234), the Alzheimer's Association (SG-20-725707), the Rainwater Charitable Foundation – The Bluefield project to cure FTD, and Global Brain Health Institute, and Agencia Nacional de Investigación y Desarrollo (ANID)/Fondo Nacional de Desarrollo Científico y Tecnológico (FONDECYT) Regular (1250091 and 1210176 and 1220995); ANID/PIA/ANILLOS ACT210096; EU Joint Programme – Neurodegenerative Disease Research (JPND-Care), DISCeRN 2025 - Health and Social Care Research with a Focus on the Moderate and Late Stages of Neurodegenerative Diseases; FONDEF ID20I10152 and ANID/FONDAP 15150012; a Wellcome Trust award for BRAIN-CLIMA: Investigating the Combined Impact of Heat and Air Pollution on Blood-Brain Barrier Integrity and Brain Aging in Latin America (335293/Z/25/Z); the Wellcome Leap CARE Program (CARE-2025-0883490149) for the project "Advancing Female-Specific Predictive Models and Risk Assessment Tools for Alzheimer's Disease in the US and Latin America; and CliCBrain (Horizon ID, 101236426; Marie Skłodowska-Curie Actions). C.D.-A. is supported by is supported by grants from ReDLat2, supported by Fogarty International Center, the NIH, the US National Institute on Aging (R01 AG057234, R01 AG075775, R01 AG21051, R01 AG083799 and CARDS-NIH), the Alzheimer's Association (SG-20-725707), the Rainwater Charitable Foundation – The Bluefield project to cure FTD, and Global Brain Health Institute, ANID/FONDECYT Regular (1210622 and 1250091), ANID/NAM22I0007, the Alzheimer's Association (AARGD-24-1310017), ANID/FOVI240065, ANID/Proyecto Exploracion 13240170, the Wellcome Leap CARE Program (CARE-2025-0883490149), CliCBrain (Horizon ID, 101236426) and A2026019S BrightFocus Foundation Standard Award Program in Alzheimer's Disease Research. J.S.Y. receives funding from the NIH National Institute on Aging (R01AG057234, P30AG062422, P01AG019724, and U19AG079774); the NIH National Institute of Neurological Disorders and Stroke (U54NS123985); the Rainwater Charitable Foundation; the Alzheimer's Association; the Global Brain Health Institute; and the Mary Oakley Foundation. The contents of this publication are solely the responsibility of the authors and do not represent the official views of these institutions, including the NIH. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Competing interests

J.S.Y. serves on the scientific advisory board for the Epstein Family Alzheimer's Research Collaboration, the Charleston Conference on Alzheimer's Disease and Taudia, and is the editor in chief of *npj Dementia*. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41591-026-04433-3.

the Multi-Partner Consortium to Expand Dementia Research in Latin America (ReDLat)

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